Introduction To Lid Disease

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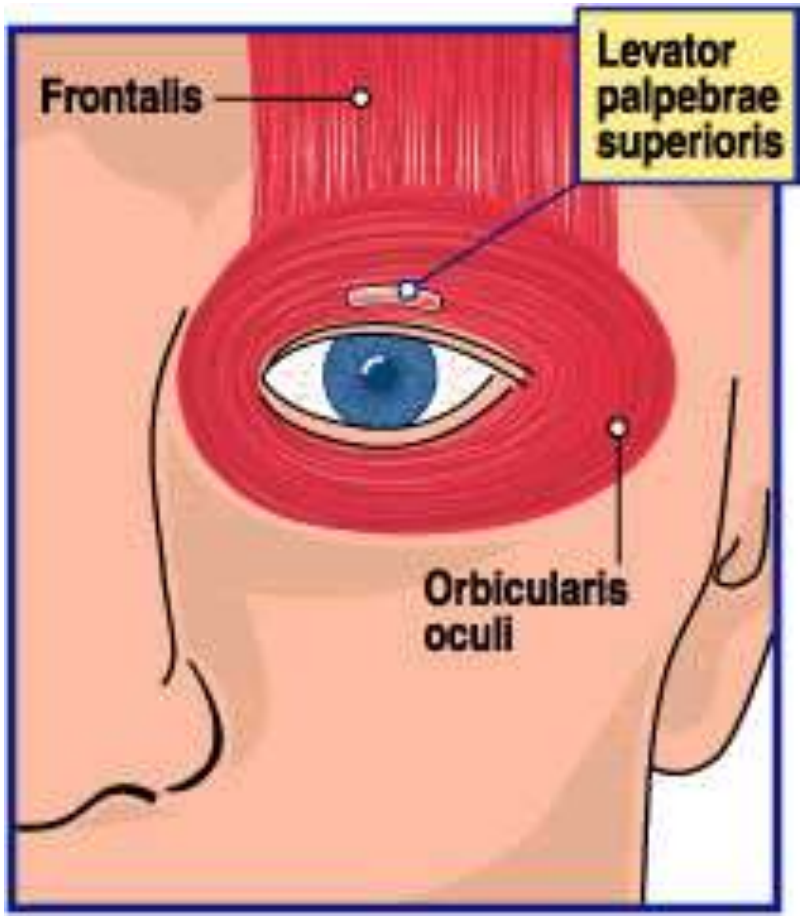
SOAP Format

- S = Subjective
  - AKA symptoms (dry eye)
  - Everything the patient says
- O = Objective
  - AKA signs
  - All the doctors observes and all the test results
- A = Assessment
  - AKA diagnosis (or diagnoses) (impression)
  - What is wrong with the patient
- P = Plan
  - For each diagnosis, what is going to be done to help the patient

Lid Anatomy

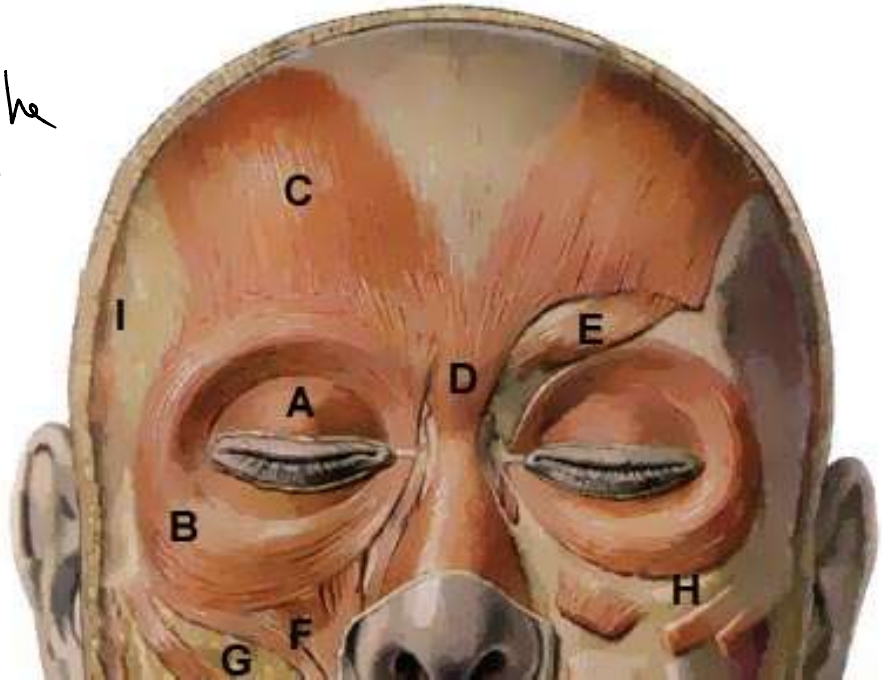
Muscles

- **Levator** lifts and open its
  striated m.
- **Mueller's muscle**
  smooth m. sup. inf. opens and keep open lid.
- **Orbicularis**
  "When you squit eyes tightly
  "Straicted" m.



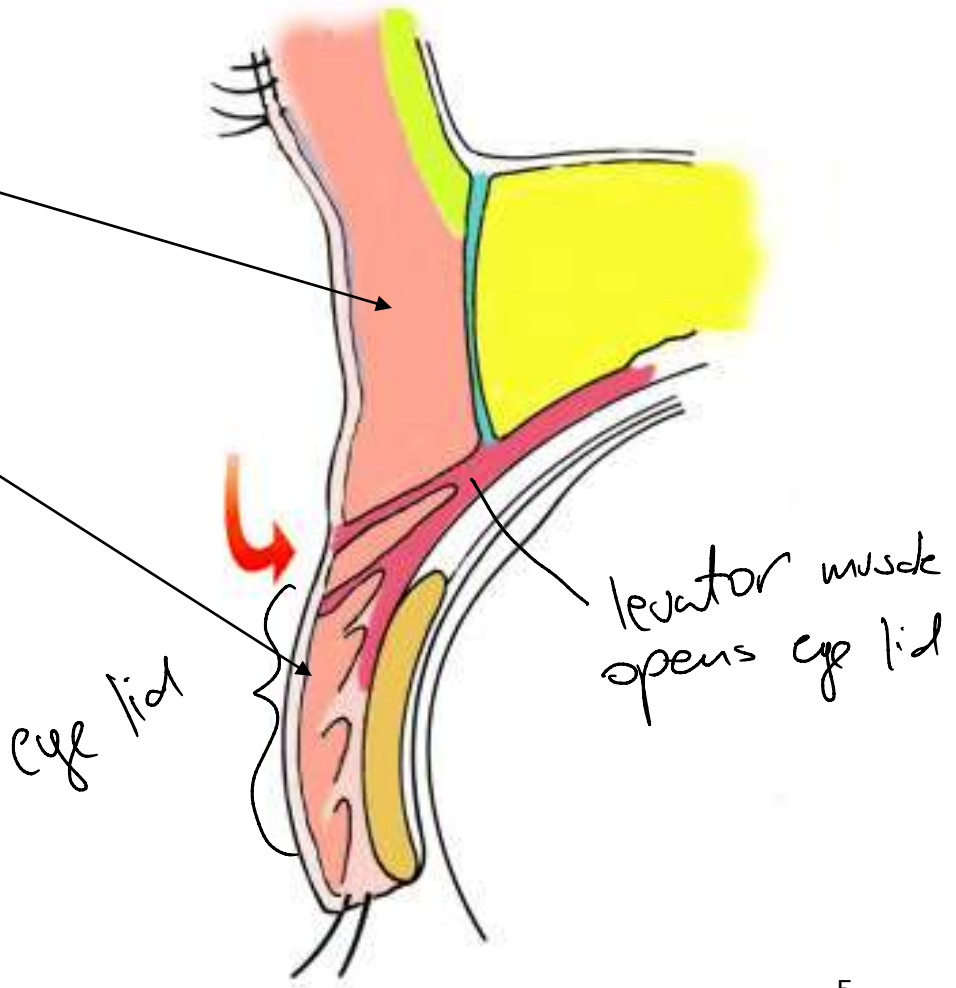
Lid Anatomy

- **A = orbicularis muscle (palpebral portion)** keeps the eye "safe"
  - **Striated**
  - **Eyelid closing muscle - used in blinking and voluntary winking**
  - **Keeps lid tight to globe**
- **B = orbicularis muscle (orbital portion)**
  - **Striated**
  - **Eyelid closing muscle - used in forced closure**



Lid Anatomy

- Orbicularis - orbital portion
- Orbicularis - palpebral portion

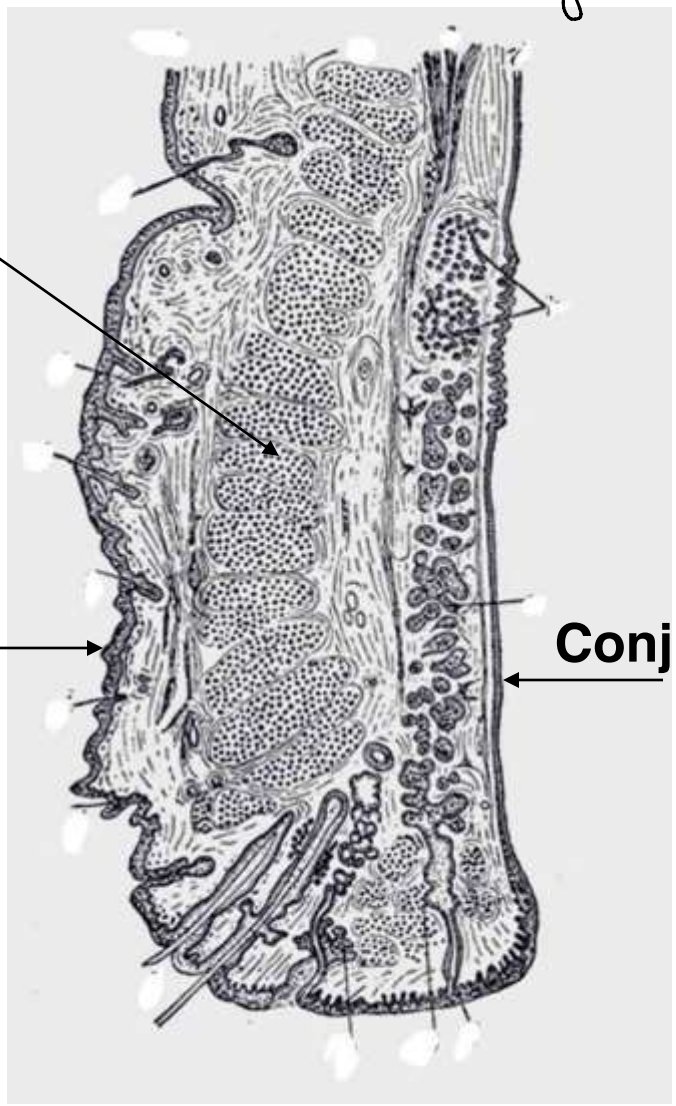


Lid Anatomy

- **Orbicularis**
  - **Innervation is from the facial nerve (CN VII)**

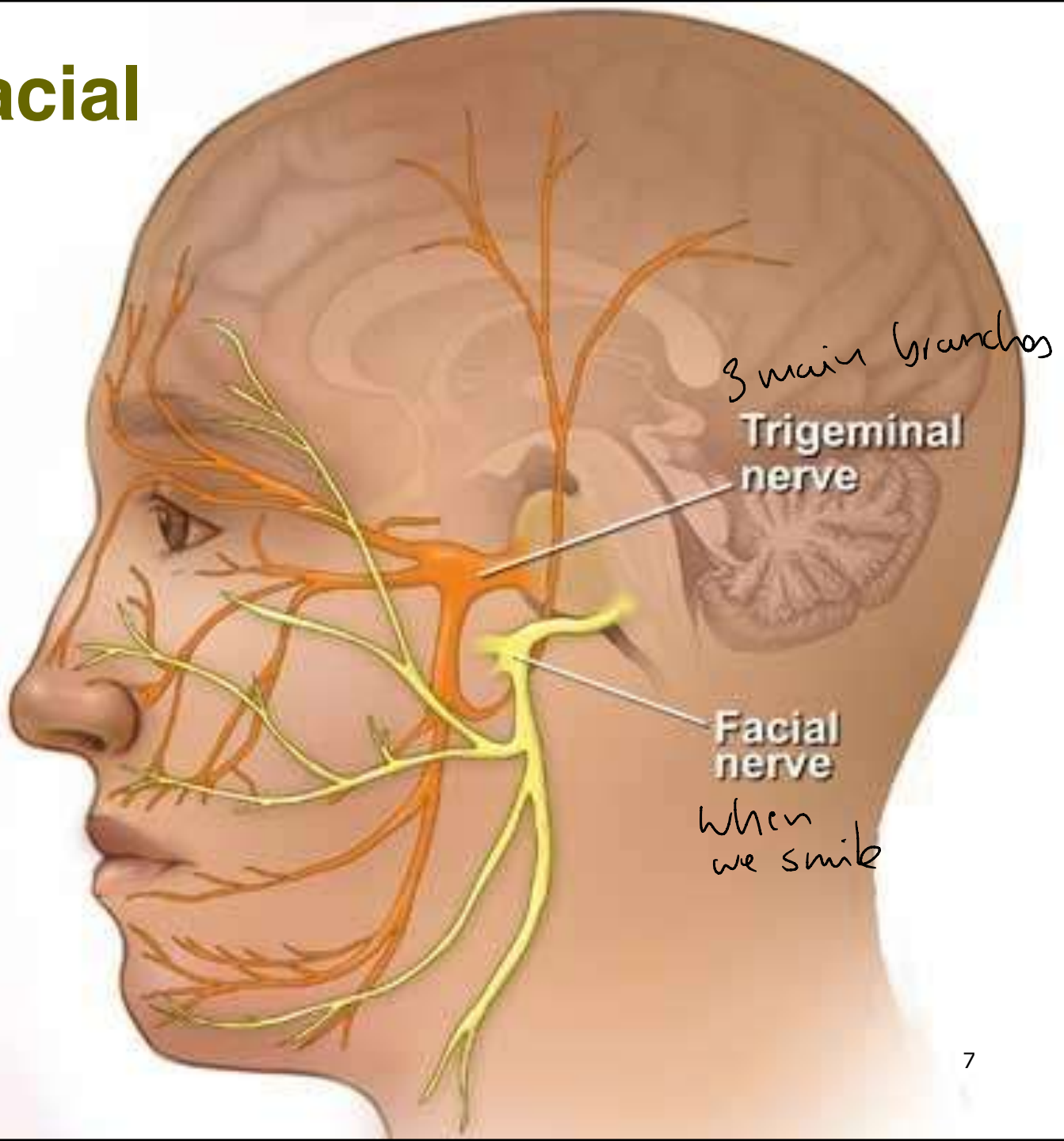
facial nerve closes our eye lid

cross section of



CNVII = Facial Nerve

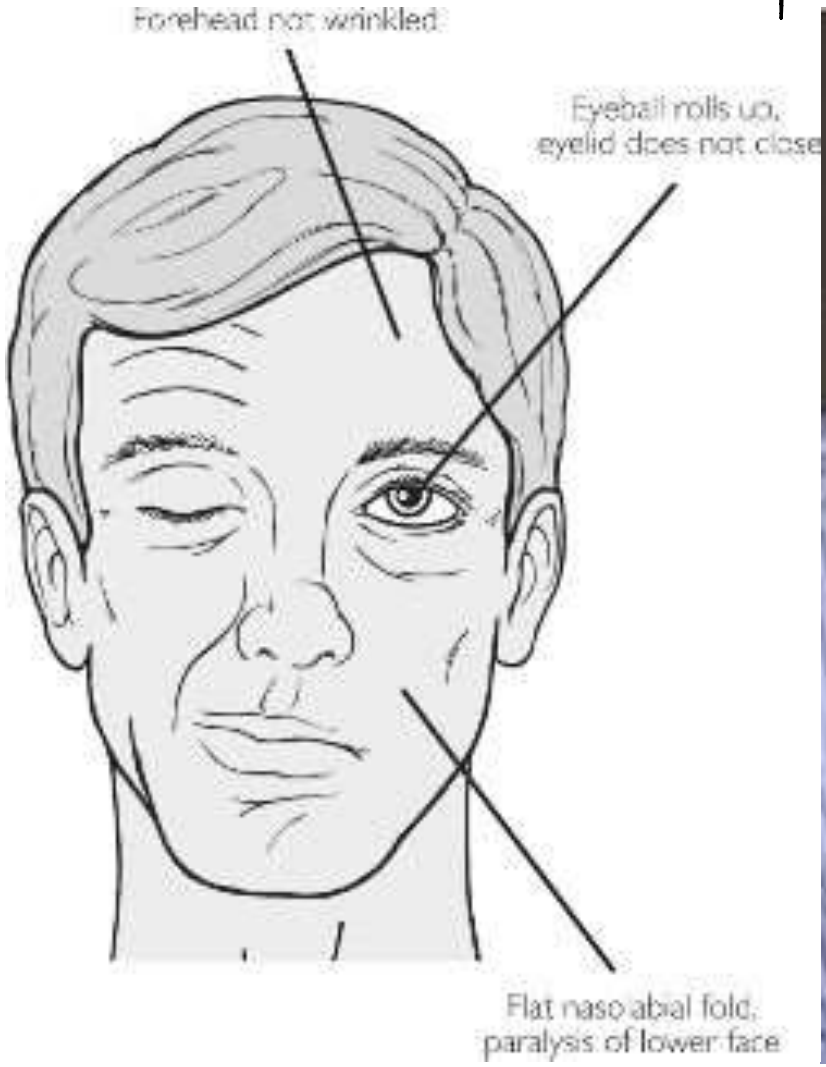
- CNVII causes:
- eyelids to close
- mouth and cheeks to ‘smile’
- forehead to ‘furrow’



Bell's Palsy

half of the face is not working (zoster, stroke dental proc. trauma) car accident

most common cause: idiopatic (don't know what it is)

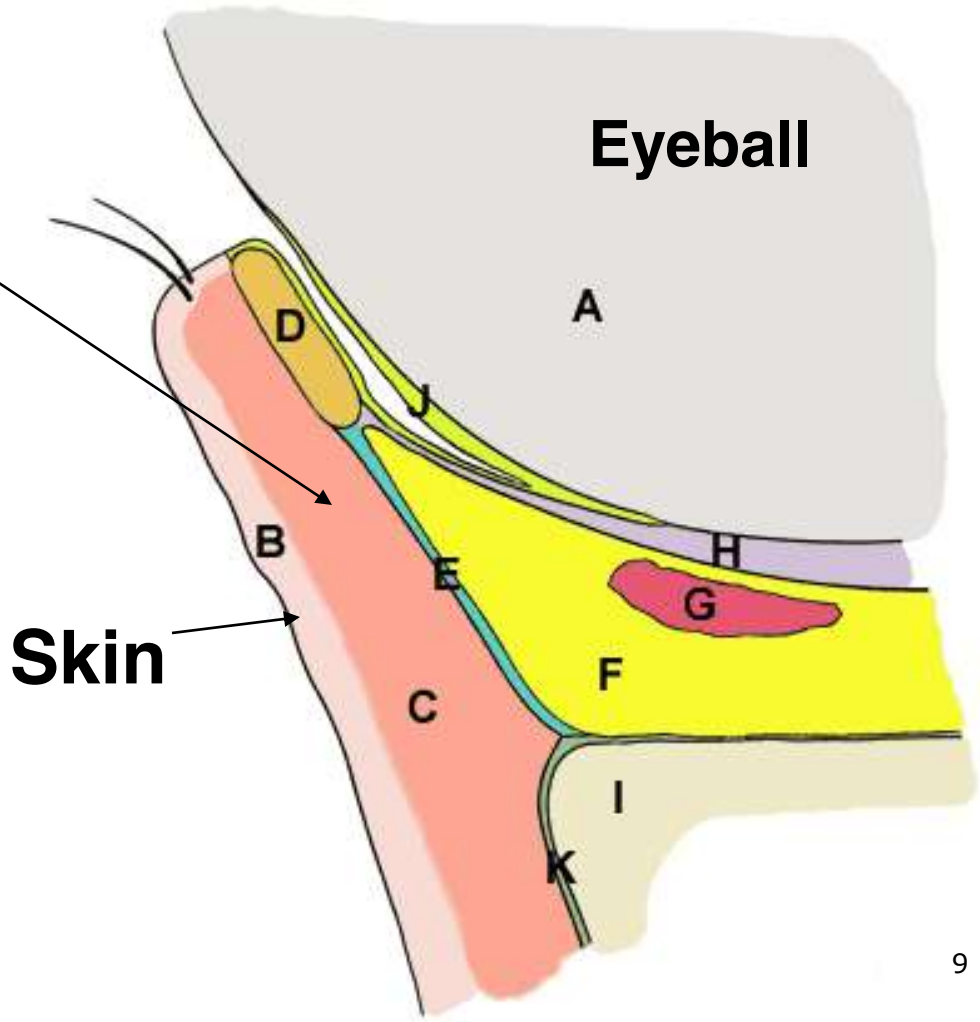




Lid Anatomy

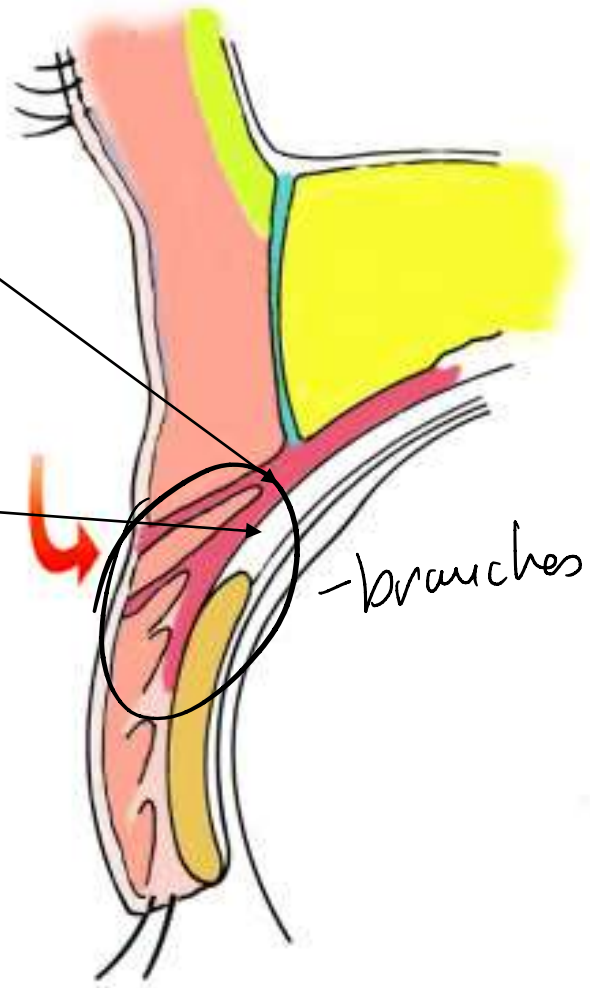
- **Orbicularis (lower lid)**

closes the eye lid from bellow



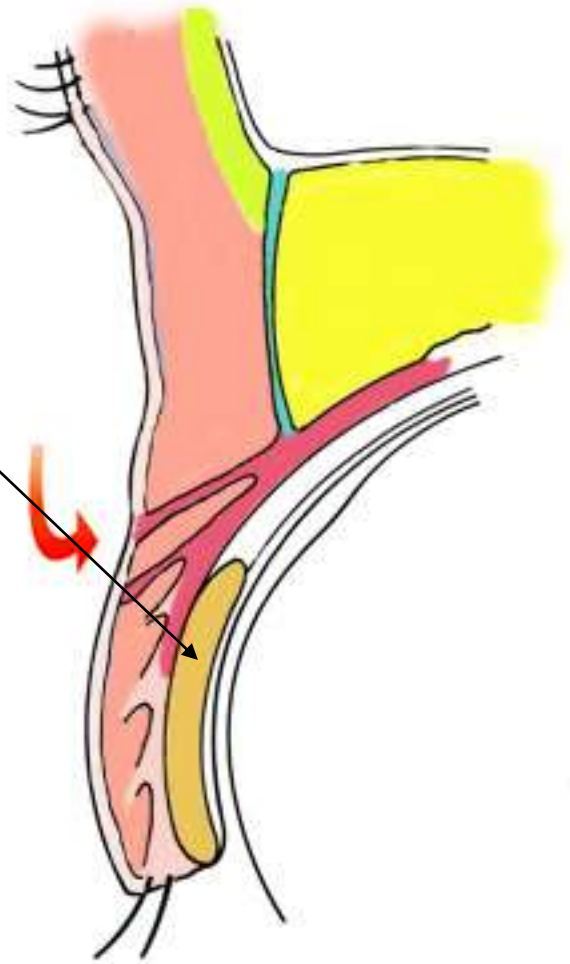
Lid Anatomy

- **Levator aponeurosis**
  - Striated
  - Lid opening muscle
  - Innervated by CN III (down and out)
  - Fibers pass through the orbicularis to attach to the skin
- **Muller's muscle**
  - Originates from the underside of the levator
  - Smooth
  - Lid opening muscle
  - Innervated by the sympathetic system
  - Inserts into superior tarsus



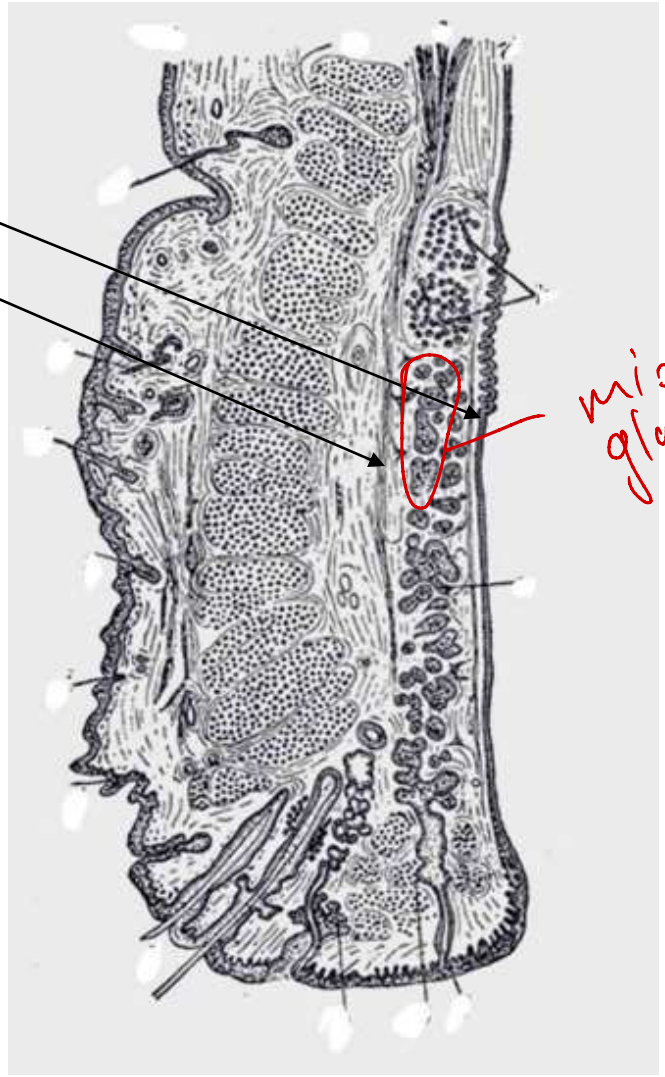
Lid Anatomy

- **Tarsal plate**
  - Dense fibrous connective tissue
  - Responsible for structural integrity of the lids
  - Upper plate about 1 cm high x 2.5 cm wide
  - Lower tarsal plate smaller dimensions



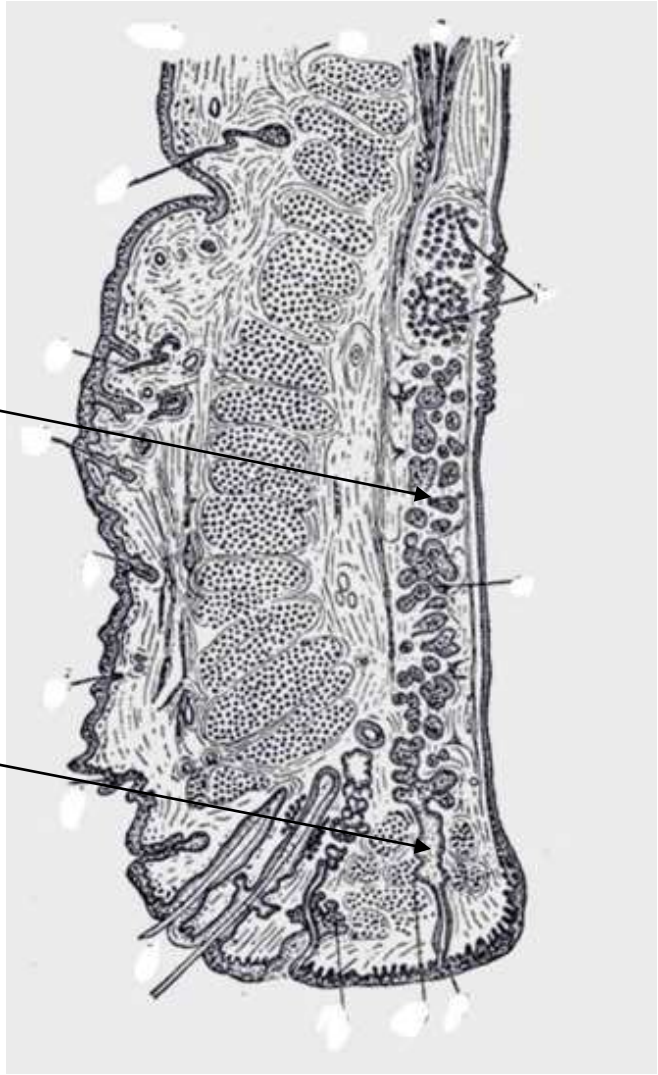
Lid Anatomy

- **Tarsal plate (between arrows)**



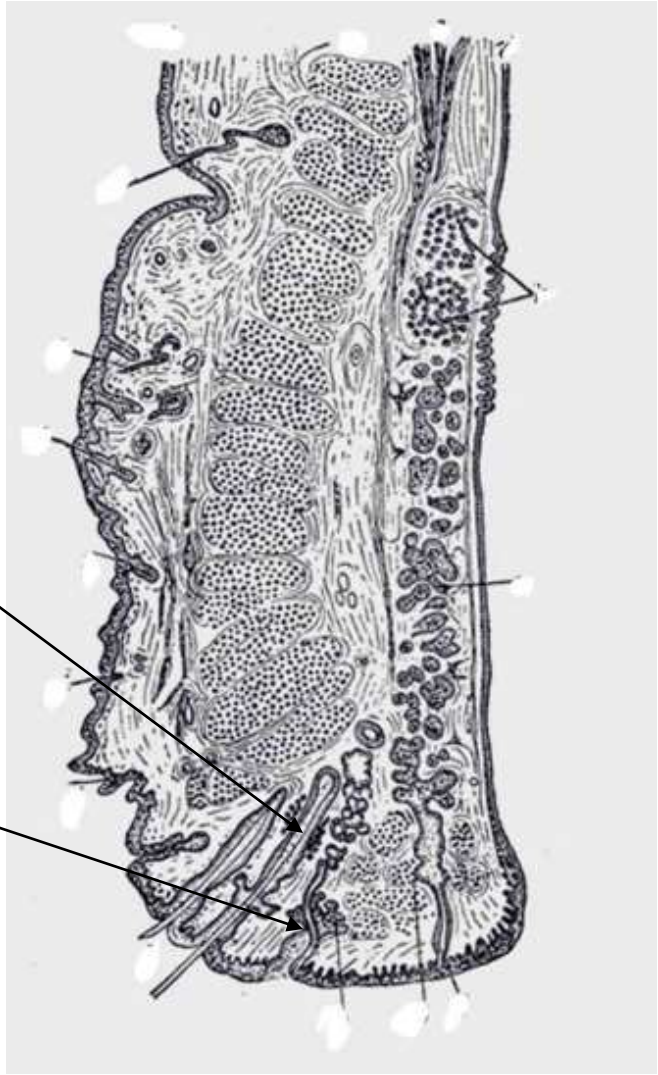
Lid Anatomy

- Each tarsus encloses about 25 meibomian glands
- Meibomian gland
  - Sebaceous gland (oil)
  - Provides the oily layer of the tear film
  - Meibomian gland opening just behind the lashes



Lid Anatomy

- Lash follicle and associated sebaceous gland(s) (Zeiss gland)
- Moll gland - sweat



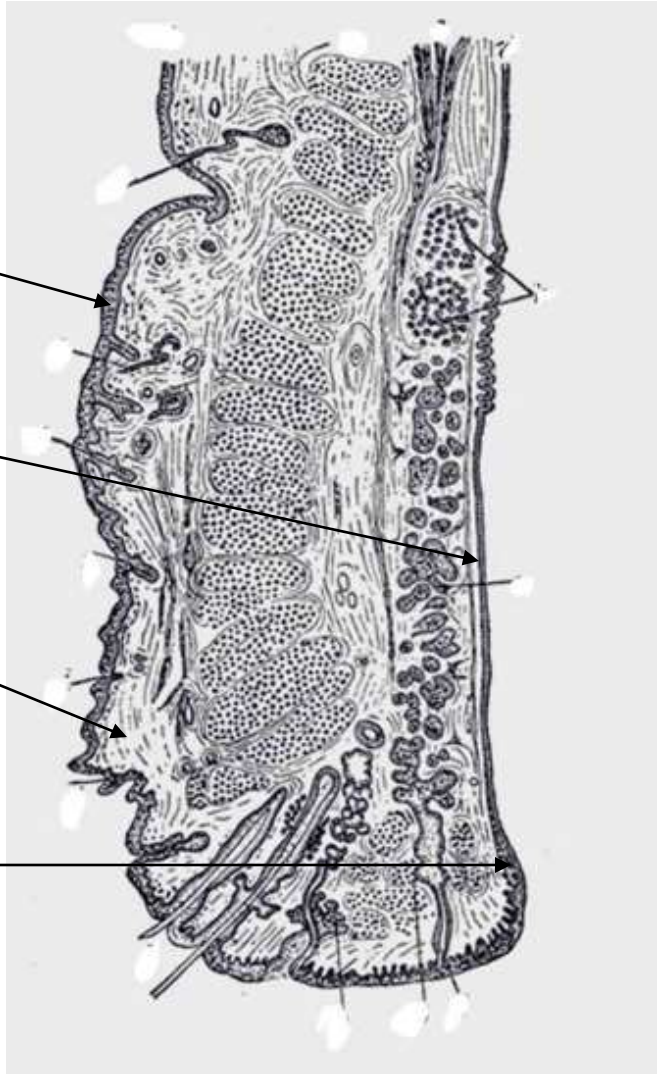
Lid Anatomy

Skin

Conjunctiva

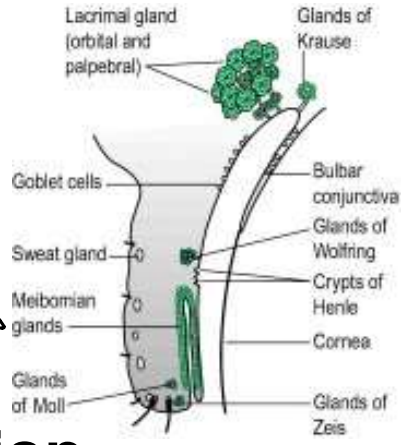
Loose connective tissue dermis

Mucocutaneous junction / gray line



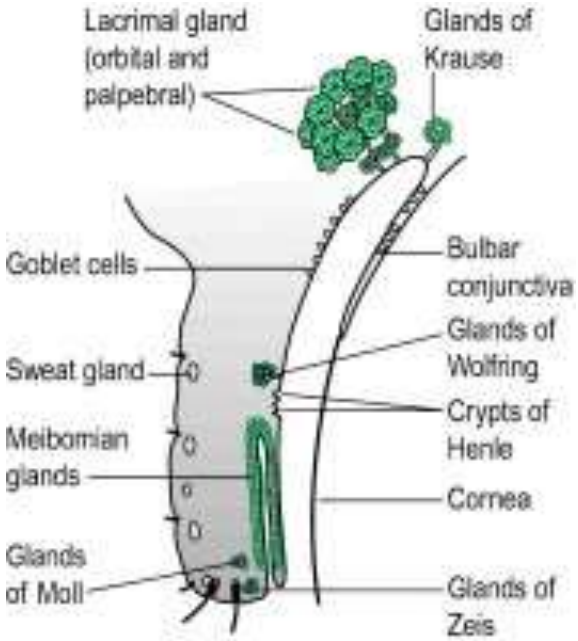
Eyelid Glands – Conjunctival Side

- **Goblet cells** toward fornix
  - In conjunctival epithelium
  - Secrete mucin
  - Abundant at plica semilunaris (caruncle) pink tissue in eye
  - None on limbus or mucocutaneous junction
- **Accessory glands of Krause and Wolfring**
  - Krause – near fornix; basal aqueous secretion
  - Wolfring – near tarsus; basal aqueous secretion water of our tears
  - Both assist lacrimal gland in aqueous secretion
  - Only lacrimal gland has reflex secretion (tearing) most of our tears?



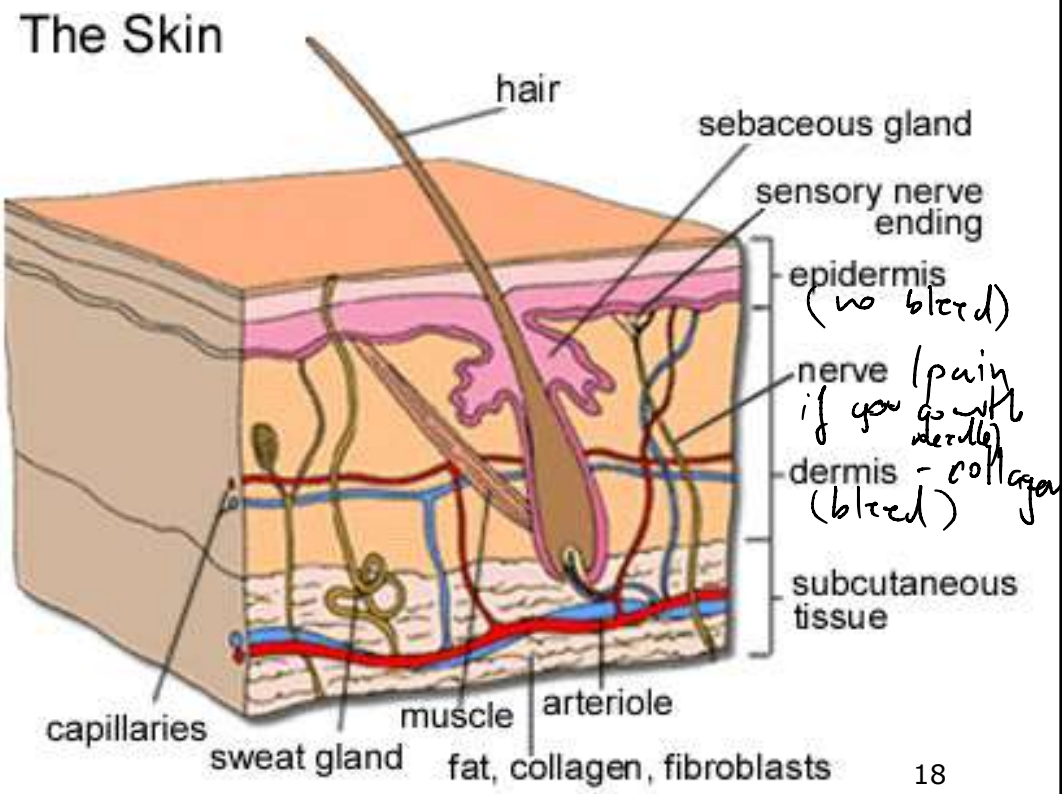
Eyelid Glands – Lid Margin and Skin

- **Meibomian glands**
  - Secrete sebum (lipid)
- **Gland of Zeiss**
  - Unilobular gland
  - Produce oils for the lash follicles
- **Gland of Moll**
  - aka ciliary gland
  - Anterior to Zeiss glands
  - Produce oils as well
- **Sweat gland**
  - aka sudoriferous gland

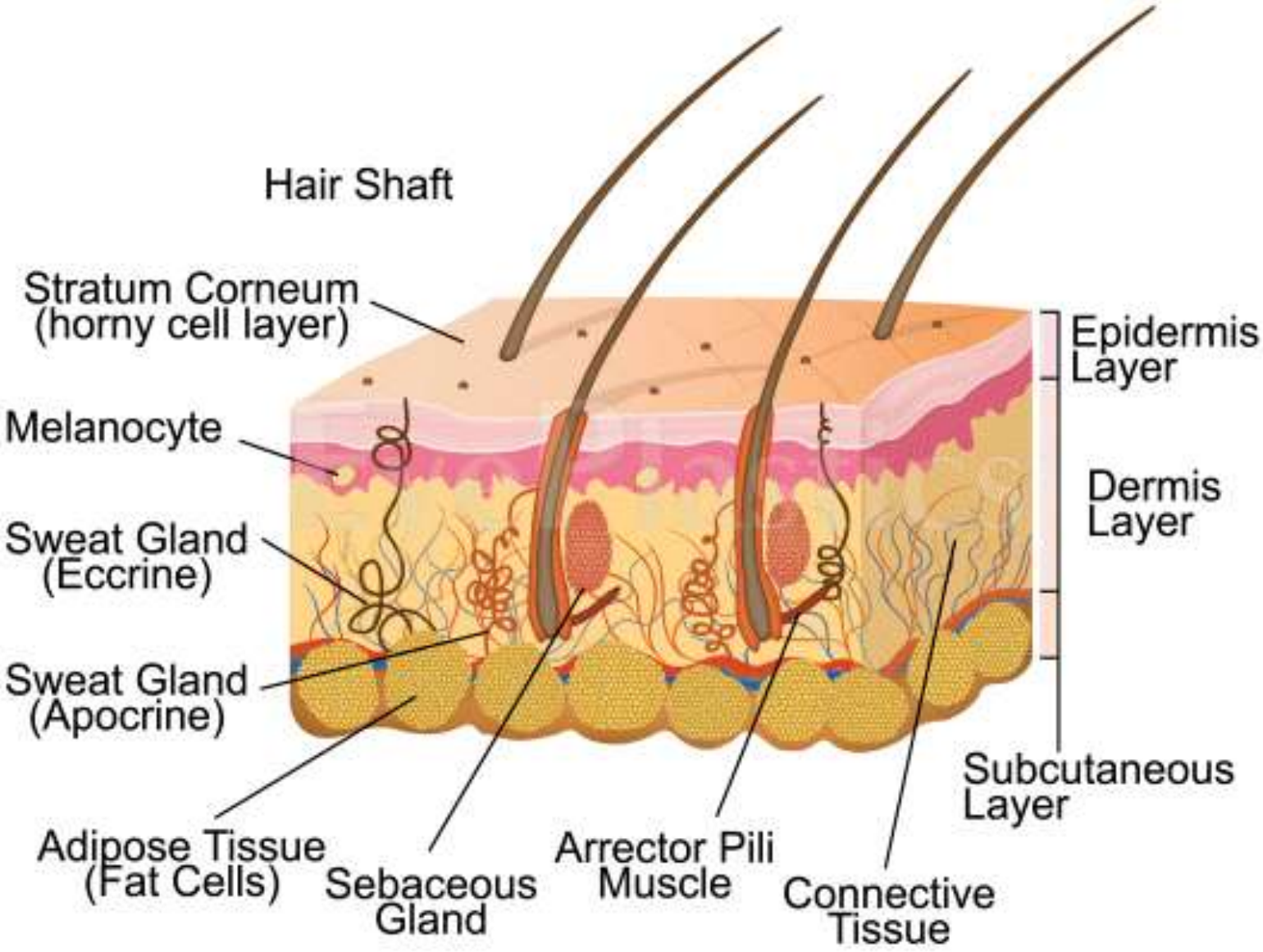


Lid Anatomy - Skin – the ~~thinner~~ thinnest part of the skin

- **Eyelid skin is the thinnest of the body**
  - **Epidermis**
  - **Dermis**
  - **Hair follicle and associated sebaceous gland**
  - **Sweat gland**
  - **Blood supply**

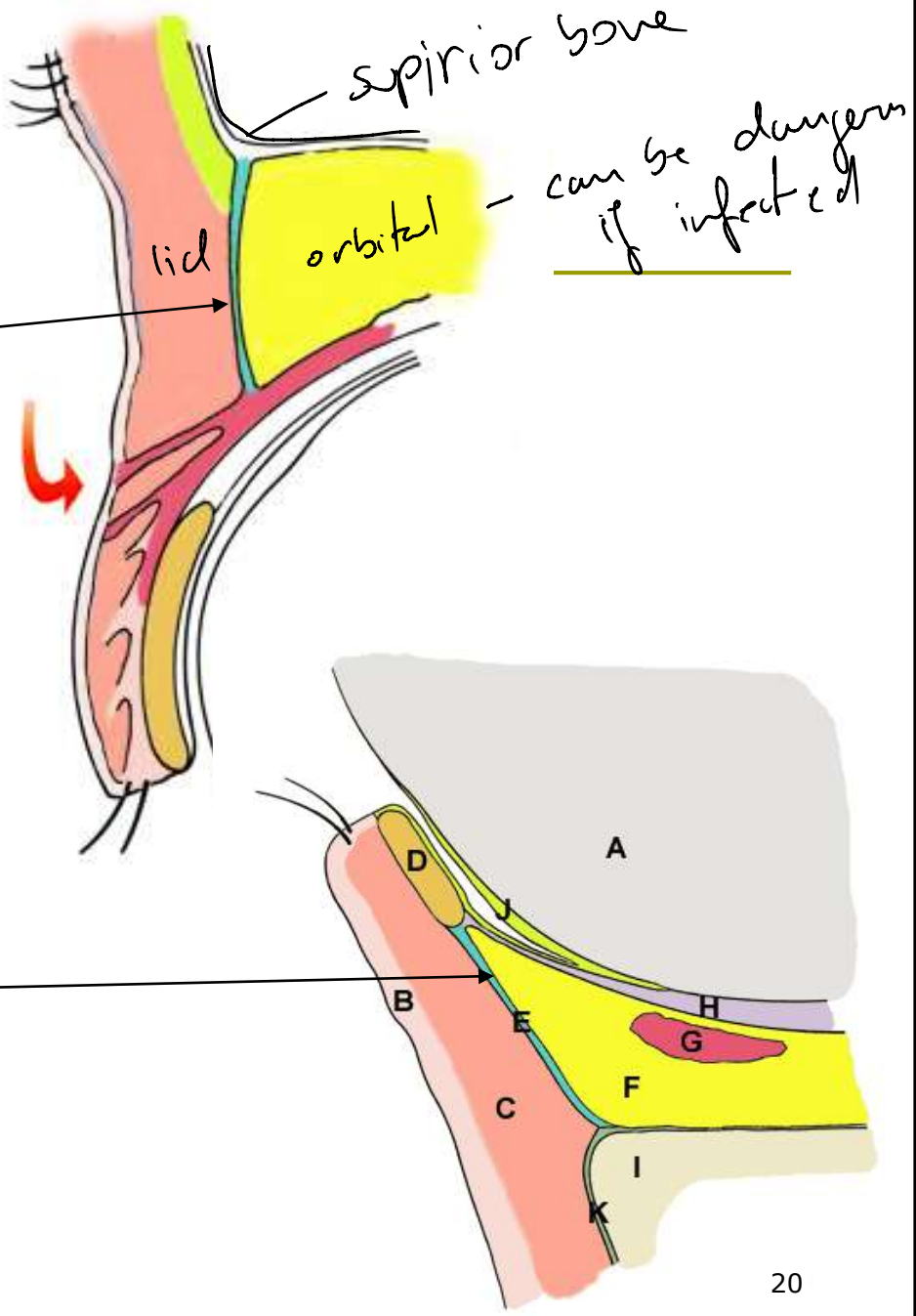


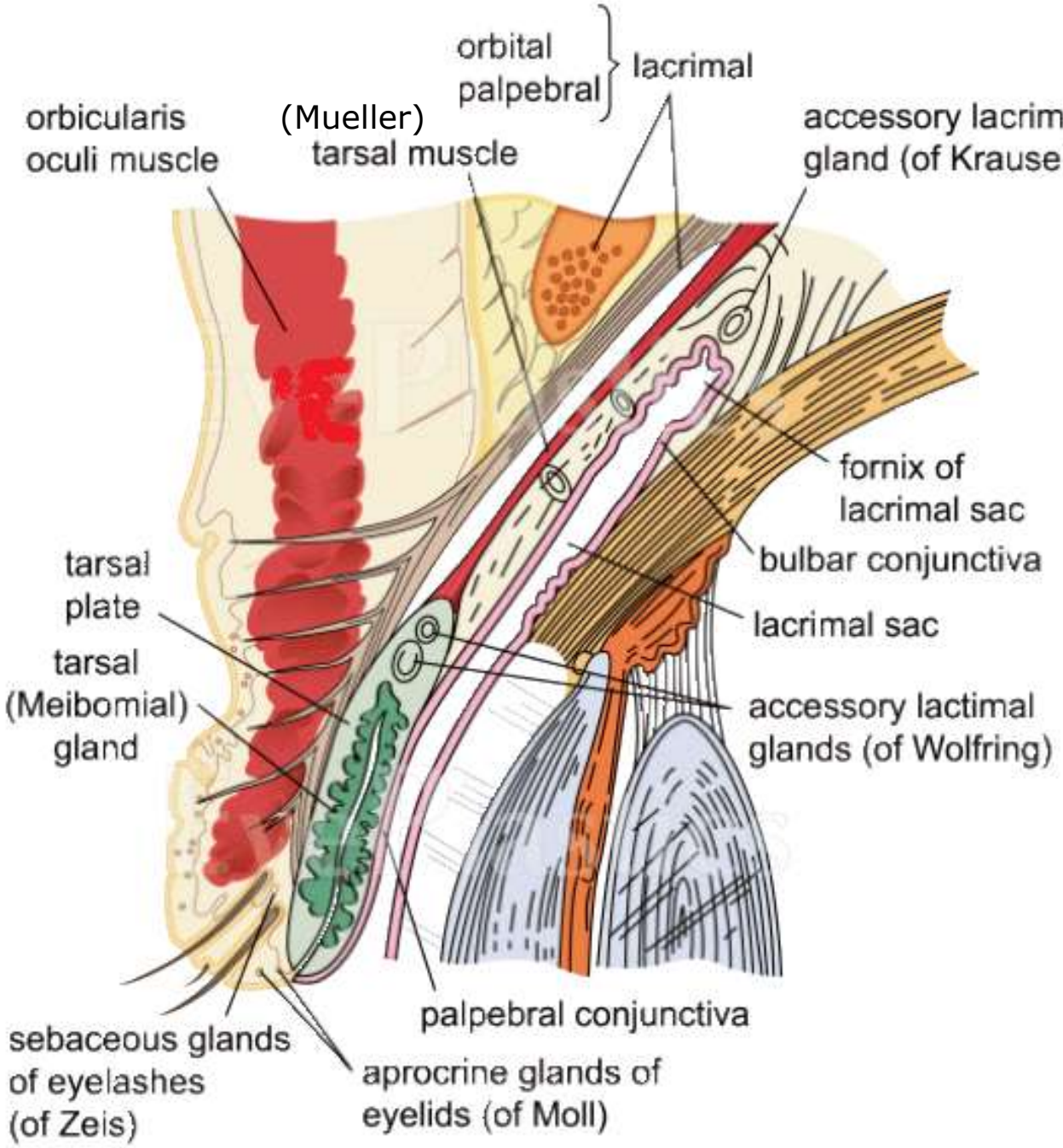
Skin Anatomy



Lid Anatomy

- **Orbital septum**
  - **Attaches peripherally at the periosteum at the orbital margin**
  - **Centrally, it fuses with levator aponeurosis (upper lid) and the tarsal plate (lower lid)**
  - **Acts as diaphragm to separate orbital structures from lid structures**





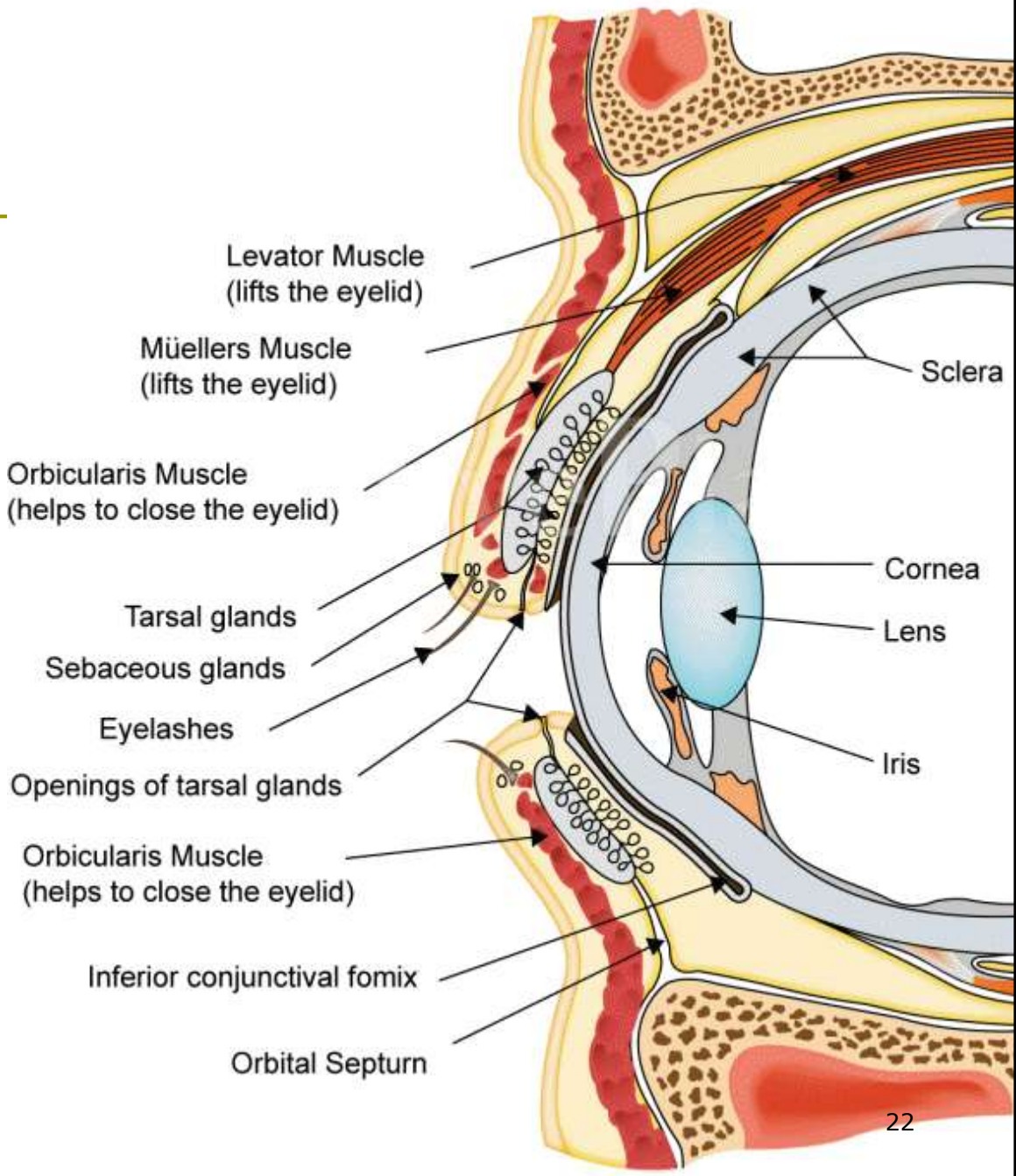
Eyelids are complex!

- Skin
- Hair
- Fascia – connective tissue
- Muscle
  - Striated
  - Smooth
- Glands meibomian Zeiss...
- Nerves
- Conjunctiva back side
- Connective tissue

...lots of structures for disease to occur!

Lid Diseases

- Position/shape
- Inflammation
- Allergy
- Infection
- Cancer
- Autoimmune
- Gland dysfunction
- Neurological
- Trauma



THANK YOU!